4.6 (a)

① $J_n(-x_p) = \frac{eD_n n_{p0}}{L_n}\left[\exp\left(\frac{eV_F}{k_BT}\right)-1\right]$

则电子反向饱和电流密度：$J_{ns} = \frac{eD_n n_{p0}}{L_n}$

$n_{p0} = \frac{n_i^2}{N_A}$，$L_n = \sqrt{D_n \tau_n}$

则 $J_{ns} = \frac{1.6\times10^{-19}\times 25 \times \frac{(1.5\times10^{10})^2}{5\times10^{15}}}{\sqrt{25 \times 0.4\times10^{-6}}} = 5.69\times10^{-11}\ A/cm^{-2}$

则电子反向饱和电流：$I_{ns} = J_{ns}\cdot A = 5.69\times10^{-11} A/cm^{-2}\times10^{-4}cm^2$
$= 5.69\times10^{-15} A$

$J_p(x_n) = \frac{eD_p p_{n0}}{L_p}\left[\exp\left(\frac{eV_F}{k_BT}\right)-1\right]$

则空穴反向饱和电流密度：$J_{ps} = \frac{eD_p p_{n0}}{L_p}$

$p_{n0} = \frac{n_i^2}{N_D}$，$L_p = \sqrt{D_p \tau_p}$

则 $J_{ps} = \frac{1.6\times10^{-19}\times 10 \times \frac{(1.5\times10^{10})^2}{1\times10^{15}}}{\sqrt{10\times0.1\times10^{-6}}} = 3.6\times10^{-10}\ A/cm^{-2}$

则空穴反向饱和电流：$I_{ps} = J_{ps}\cdot A = 3.6\times10^{-10} A/cm^{-2}\times10^{-4}cm^2$
$= 3.6\times10^{-14} A$

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② $\Delta p_n(x) = p_n(x) - p_{n0} = p_{n0}\left[\exp\left(\frac{eV_F}{k_BT}\right)-1\right]\exp\left(\frac{x_n - x}{L_p}\right)$, $x \geq x_n$

$x = x_n$, $p_n(x_n) - p_{n0} = p_{n0}\left[\exp\left(\frac{eV_F}{k_BT} - 1\right)\right]$

即 $p_n(x_n) = p_{n0}\exp\left(\frac{eV_F}{k_BT}\right) = p_{n0}\exp\left(\frac{e\frac{1}{2}V_D}{k_BT}\right)$

$$V_D = V_T \cdot \ln\left(\frac{N_A \cdot N_D}{n_i^2}\right)$$
$$= \frac{k_BT}{e}\ln\left(\frac{N_A \cdot N_D}{n_i^2}\right)$$

则 $p_n(x_n) = p_{n0}\cdot\exp\cdot\left[\frac{e}{k_BT}\cdot\frac{1}{2}\cdot\frac{k_BT}{e}\ln\left(\frac{N_A N_D}{n_i^2}\right)\right]$

$$= p_{n0}\cdot\sqrt{\frac{N_A N_D}{n_i^2}}$$
$$= \frac{n_i^2}{N_D}\cdot\frac{\sqrt{N_D N_A}}{n_i} = \frac{n_i\sqrt{N_A}}{\sqrt{N_D}} = \frac{1.5\times10^{10}\times\sqrt{5\times10^{15}}}{\sqrt{1\times10^{15}}}$$
$$= 3.35\times10^{10}\ cm^{-3}$$

③ $\Delta p_n(x) = p_n(x) - p_{n0} = p_{n0}\left[\exp\left(\frac{eV_F}{k_BT}\right)-1\right]\exp\left(\frac{x_n - x}{L_p}\right)$,

$$V_F = \frac{1}{2}V_D = \frac{1}{2}\cdot\frac{k_BT}{e}\cdot\ln\left(\frac{N_A N_D}{n_i^2}\right)$$

则 $J_p(x_n + \frac{1}{2}L_p) = eD_p\frac{d(\Delta p_n(x))}{dx}\Big|_{x = x_n + \frac{1}{2}L_p}$

$$= \left\{eD_p\cdot p_{n0}\left[\exp\left(\frac{eV_F}{k_BT}\right)-1\right]\cdot\exp\left(\frac{x_n - x}{L_p}\right)\cdot\left(-\frac{1}{L_p}\right)\right\}\Big|_{x = x_n + \frac{1}{2}L_p}$$

$$= eD_p\cdot p_{n0}\left[\exp\left(\frac{eV_F}{k_BT}\right)-1\right]\cdot e^{-\frac{1}{2}}\left(-\frac{1}{L_p}\right)$$

$$= 3.6\times10^{-10}\times\left(\sqrt{\frac{N_A N_D}{n_i^2}} - 1\right)\cdot e^{-\frac{1}{2}}$$

$$= 3.6\times10^{-10}\times\left(\sqrt{\frac{5\times10^{30}}{(1.5\times10^{10})^2}} - 1\right)\cdot e^{-\frac{1}{2}} = 3.25\times10^{-5}\ A/cm^{-2}$$

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则 $I_p(x_n+\frac{1}{2}L_p)=J_p(x_n+\frac{1}{2}L_p)\cdot A$

$=3.25\times10^{-5}A/cm^{-2}\times10^{-4}cm^2$

$=3.25\times10^{-9}A$.

(b) ① $\Delta p(x)=p_{n0}\left[\exp\left(\frac{eV_F}{k_BT}\right)-1\right]e^{\frac{-x}{L_p}}$ $(x\geq0)$

$p_{n0}=\frac{n_i^2}{N_D}$

$L_p=\sqrt{D_p\tau_p}$, $\cancel{V_F=\frac{kT}{e}\ln\left[\frac{N_A\cdot N_D}{n_i^2}\right]}$

则 $\Delta p(x)=\frac{(1.5\times10^{10})^2}{10^{16}}\times\left[\exp\left(\frac{1.6\times10^{-19}\times0.61}{1.38\times10^{-23}\times300}\right)-1\right]\cdot e^{\frac{-x}{\sqrt{8\times0.01\times10^{-6}}}}$

$=\cancel{3.89\times10^{14}}$ $3.9\times10^{14}\cdot e^{-3.54\times10^3\cdot x}$ $(x\geq0)$

② $x=3\times10^{-4}cm$.

空穴扩散电流密度:

$J_p(x)=eD_p\cdot\frac{d(\Delta p(x))}{dx}$

$=eD_p\cdot3.9\times10^{14}\times e^{-3.54\times10^3\cdot x}\cdot\frac{1}{\sqrt{8\times0.01\times10^{-6}}}$

$=1.6\times10^{-19}\times8\times3.9\times10^{14}\times e^{-3.54\times10^3\times3\times10^{-4}}\times(3.54\times10^3)$

$=0.611\ A/cm^{-2}$.

$J_总=\left(\frac{eD_nn_{p0}}{L_n}+\frac{eD_pp_{n0}}{L_p}\right)\cdot\left[\exp\left(\frac{eV_F}{k_BT}\right)-1\right]$

$=\left(\frac{1.6\times10^{-19}\times(1.5\times10^{10})^2}{5\times10^{16}}\times\sqrt{\frac{23}{0.05\times10^{-6}}}+\frac{1.6\times10^{-19}\times(1.5\times10^{10})^2}{10^{16}}\cdot\sqrt{\frac{8}{0.01\times10^{-6}}}\right)$

$\cdot\left[\exp\left(\frac{1.6\times10^{-19}\times0.61}{1.38\times10^{-23}\times300}\right)-1\right]$

$=2.03\ A/cm^{-2}$

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则电子电流密度：

$$J_n = J_{总} - J_p$$

$$= 2.031 - 0.611 = 1.420\ A/cm^{-2}$$

4.9 (a) 肖特基二极管：

$$J_{ST} = 3\times10^{-8}\ A/cm^2$$

$$J = J_{ST}\left[\exp\left(\frac{eV}{k_BT}\right)-1\right] = 1\times10^{-3}A/5\times10^{-4} = 2A/cm^{-2}$$

则 $\frac{eV}{k_BT} = \ln\left(\frac{2}{3\times10^{-8}}+1\right) = 18.02$

故 $V = 18.02\times\frac{0.0259eV}{e}$

$$= 0.467V$$

PN结二极管：

$$J_s = 3\times10^{-12}\ A/cm^2$$

$$J = J_s\left[\exp\left(\frac{eV}{k_BT}\right)-1\right] = 1\times10^{-3}A/5\times10^{-4} = 2A/cm^{-2}$$

则 $V = \ln\left(\frac{2}{3\times10^{-12}}+1\right)\times\frac{0.0259eV}{e}$

$$= 0.705V$$

(b) 并联：两管电压相同.

肖特基：$I = I_{ST}\cdot\left[\exp\left(\frac{eV}{k_BT}\right)-1\right]$

PN结：$I = I_s\left[\exp\left(\frac{eV}{k_BT}\right)-1\right]$

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故 $\frac{I_1}{I_2}=\frac{I_{ST}}{I_S}=\frac{5\times10^{-8}A}{10^{-12}A}=5\times10^{4}$

又 $I_1+I_2=0.5\times10^{-3}A$，得 $\begin{cases} I_1=4.9999\times10^{-4}A. \\ I_2=9.9998\times10^{-9}A. \end{cases}$

$I_{总}=(I_{ST}+I_S)\left[\exp\left(\frac{eV}{k_BT}\right)-1\right]$

得 $V_1=V_2=0.0259\times\ln\left[\left(\frac{5\times10^{-4}}{5\times10^{-8}+10^{-12}}\right)+1\right]=0.239V.$

串联：$I_1=I_2=0.5mA=5\times10^{-4}A.$

肖特基：$I_1=I_{ST}\cdot\left[\exp\left(\frac{eV_1}{k_BT}\right)-1\right]$

$V_1=0.0259\times\ln\left(\frac{5\times10^{-4}}{5\times10^{-8}}+1\right)=0.239V.$

PN结：$I_2=I_S\cdot\left[\exp\left(\frac{eV_2}{k_BT}\right)-1\right]$

$V_2=0.0259\times\ln\left(\frac{5\times10^{-4}}{10^{-12}}+1\right)=0.519V.$

补充题：

作业题3：

(a) Ge是n型半导体，GaAs是p型半导体，组成异质结.

(b) Ge晶格常数：$a_1=5.657Å$

GaAs晶格常数：$a_2=5.653Å$

失配率：$\frac{2|a_1-a_2|}{a_1+a_2}=\frac{(5.657-5.633)\times2}{(5.657+5.653)}=0.07\%$

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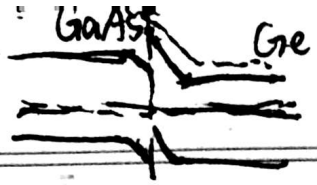

(c) $\chi_{GaAs} - \chi_{Ge} = 4.07eV - 4.01\,eV = 0.06\,eV$.

(d) $(\chi_{GaAs} + Eg_{GaAs}) - (\chi_{Ge} + Eg_{Ge})$

$= (4.07eV + 1.43eV) - (4.01eV + 0.66eV)$

$= 0.83eV$.

(e) GaAs：p型半导体，且 $N_A = 8\times10^{15} \gg n_i = 1.8\times10^{6}$

$E_F - E_{V0} = -k_BT\cdot\ln\left(\frac{N_A}{N_V}\right) = -0.0259\times\ln\left(\frac{8\times10^{15}}{6\times10^{18}}\right)$

$= 0.17eV$.

(f) Ge：n型半导体，且 $N_D = 4.5\times10^{15} \gg 2.4\times10^{13}$

$E_C - E_F = -k_BT\ln\left(\frac{N_D}{N_C}\right)$

$= -0.0259\times\ln\left(\frac{4.5\times10^{15}}{1.04\times10^{19}}\right) = 0.20eV$

(g) $\Delta E_0 = (\chi_{GaAs} + Eg_{GaAs} - 0.17) - (\chi_{Ge} + 0.2)$

$= 4.07 + 1.43 - 0.17 - 4.01 - 0.2$

$= 1.12eV$

(h) GaAs处：$x_p = \left(\frac{2\varepsilon_n\varepsilon_p N_D V_D}{eN_A(\varepsilon_n N_D + \varepsilon_p N_A)}\right)^{\frac{1}{2}}$

$= \left(\frac{2\times13.1\times8.854\times10^{-14}\times16\times4.5\times10^{15}\times1.12}{1.6\times10^{-19}\times8\times10^{15}\times(16\times4.5\times10^{15}+13.1\times8\times10^{15})}\right)^{\frac{1}{2}}$

$= 2.87\times10^{-5}cm$

Ge处：$x_n = \left(\frac{2\varepsilon_n\varepsilon_p N_A V_D}{eN_D(\varepsilon_n N_D + \varepsilon_p N_A)}\right)^{\frac{1}{2}} = 5.11\times10^{-5}cm$

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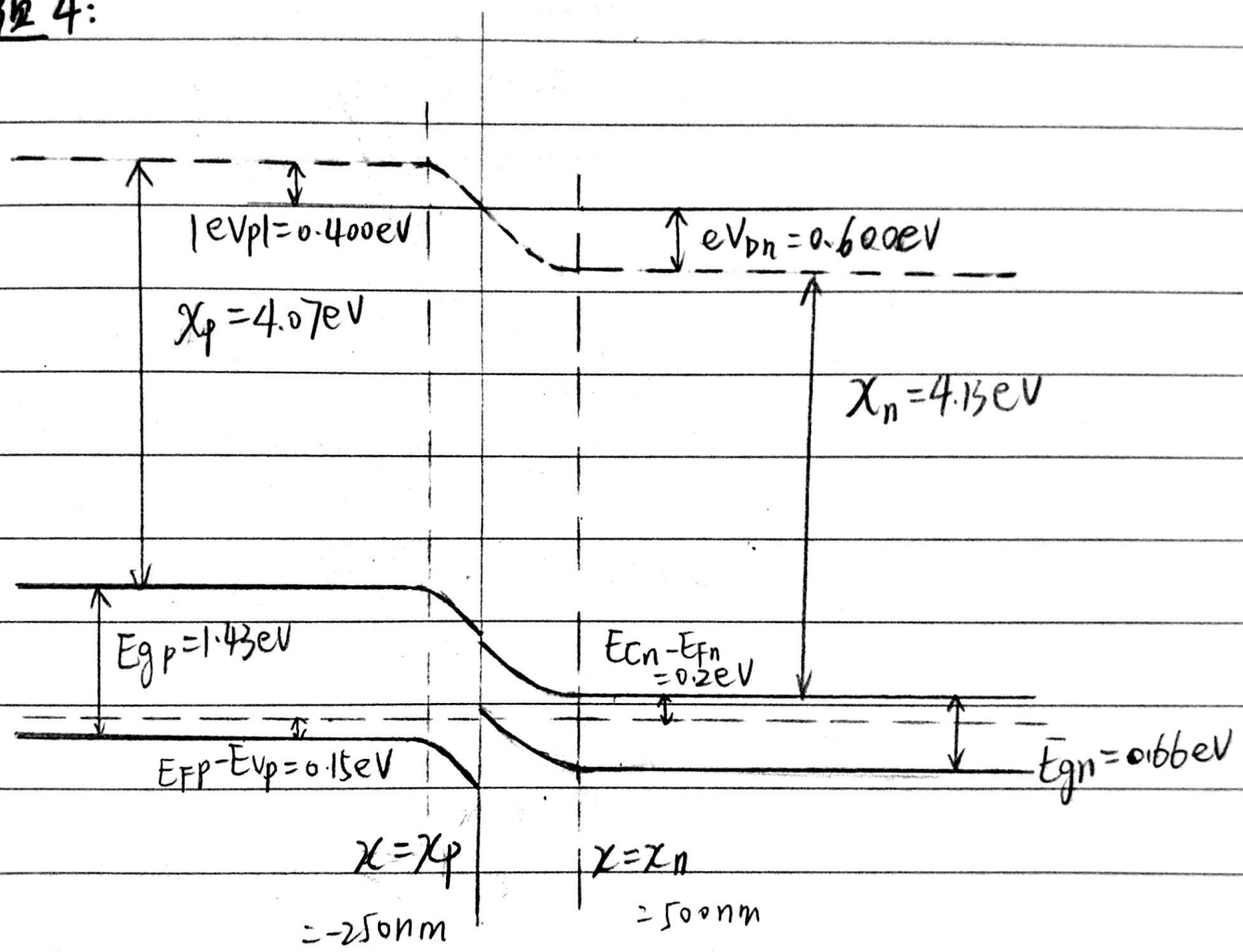
Ge处:

(i) $V_{Dn} = \dfrac{eN_D x_n^2}{2\varepsilon_n} = \dfrac{1.6\times10^{-19}\times4.5\times10^{15}\times(5.11\times10^{-5})^2}{2\times8.854\times10^{-14}\times16} = 0.664\,V$

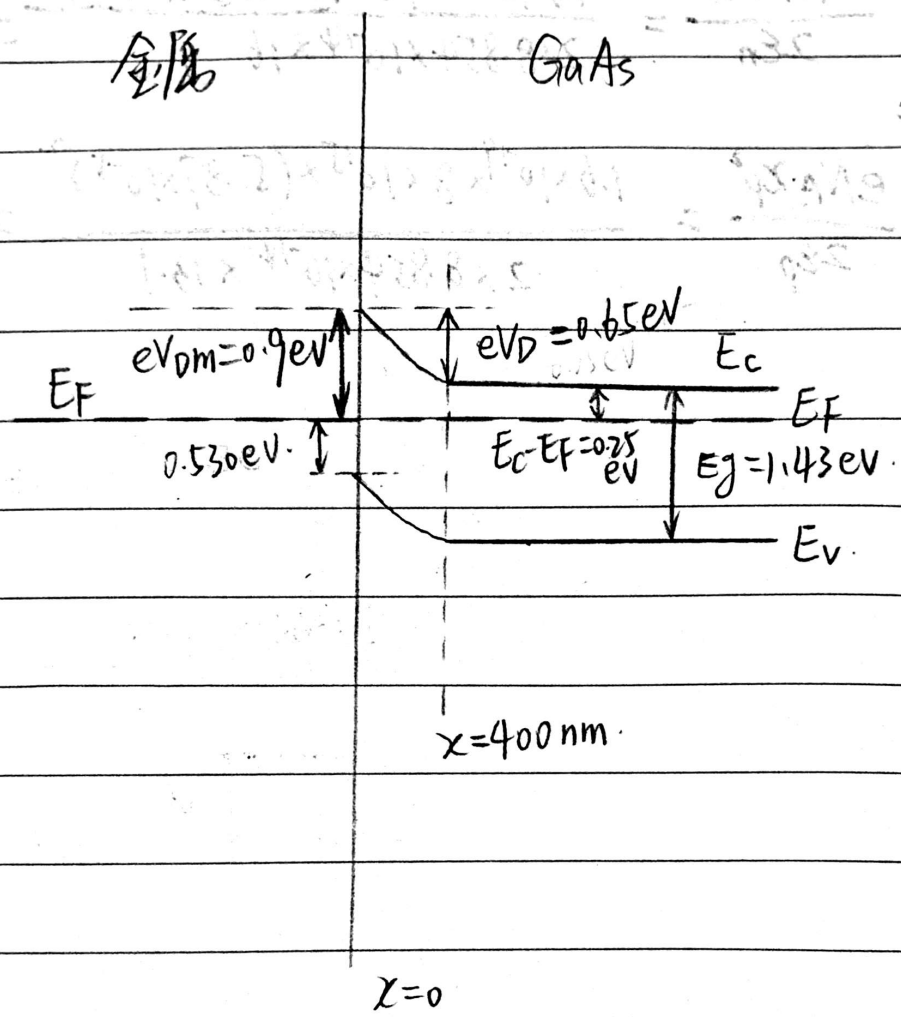
GaAs处:

$V_{Dp} = \dfrac{eN_A x_p^2}{2\varepsilon_p} = \dfrac{1.6\times10^{-19}\times8\times10^{15}\times(2.87\times10^{-5})^2}{2\times8.854\times10^{-14}\times13.1} = 0.454\,V.$

作业题4:



作业题5:


金属
GaAs
$eV_{Dm}=0.9eV$
$eV_D=0.65eV$
E_c
E_F
E_F
0.530eV
$E_c-E_F=0.25eV$
$E_g=1.43eV$
E_v
$x=400nm$
$x=0$